📄 PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project: SIP & MOVE (KSRTC Smart Beverage Ordering System)

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1. 🧠 Product Overview

SIP & MOVE is an onboard digital ordering system designed for KSRTC buses, allowing passengers to order tea/coffee through in-seat screens. Orders are processed via a centralized backend and fulfilled manually by onboard staff using a vending machine.

The system ensures:

- Seamless passenger ordering experience
- Efficient order management for staff
- Controlled access to vending machine (no passenger interaction)

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2. 🎯 Objective

Build a **Minimum Viable Product (MVP)** consisting of:

1. Passenger-facing web UI (runs on bus screens)
2. Firebase backend for real-time order handling
3. Staff panel for managing and fulfilling orders

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3. 👥 Users

3.1 Passenger

- Located in bus seat
- Uses in-seat screen
- Can place orders only

3.2 Staff

- Bus staff handling orders
- Uses mobile/tablet
- Prepares and delivers beverages

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4. 🔄 System Flow

1. Passenger selects item on screen
2. Passenger confirms order
3. Order stored in Firebase
4. Staff panel receives real-time order
5. Staff prepares item using vending machine
6. Staff delivers to passenger seat
7. Staff marks order as "Delivered"

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5. 🧩 Features & Requirements

5.1 Passenger Web UI

Functional Requirements

- Display menu:
  - Tea
  - Coffee
- Optional customization:
  - Sugar level (Low / Medium / High)
- Show:
  - Seat number (pre-configured or input once)
- Actions:
  - Place order
  - View confirmation screen

UI Screens

Screen 1: Menu

- Buttons:
  - Tea

    - Coffee
- Sugar selection
- “Order Now” button

Screen 2: Confirmation

- Message:
    - “Order Confirmed”
    - Seat Number
    - Estimated delivery time (static: ~3 minutes)

Non-Functional Requirements

- Must load instantly (<2 sec)
- Touch-friendly UI
- Works on low-power embedded screens
- Runs in browser (no app install)

5.2 Staff Panel

Functional Requirements

- Login (simple PIN or no-auth for MVP)
- View orders:
    - Seat number
    - Item (Tea/Coffee)
    - Sugar level
    - Timestamp
- Order status:
    - Pending
    - Preparing
    - Delivered
- Ability to update status

Behavior

- Orders appear in real-time
- Sorted by time (oldest first)

5.3 Firebase Backend

Services Used

- Firebase Realtime Database (or Firestore)
- Firebase Hosting (optional)

Data Structure (Example)

```
orders: {
  order_id_1: {
    seatNumber: "12",
    item: "Tea",
    sugar: "Medium",
    status: "Pending",
    timestamp: 1710000000
  }
}
```

Functional Requirements

- Store all orders
- Real-time sync between:
    - Passenger UI
    - Staff panel
- Update order status dynamically

Security Rules (MVP)

- Open read/write (for testing phase)

- Later: restrict access

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6. ⚙️ Technical Stack

Frontend

- HTML / CSS / JavaScript
  (Optional: React if scalability needed)

Backend

- Firebase

Hosting

- Firebase Hosting OR local hosting

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7. 🔌 Connectivity

MVP Mode:

- Internet-based (Firebase)

Future:

- Local WiFi server (Raspberry Pi)

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8. 📦 Hardware Integration Scope

Included:

- None (manual operation of vending machine)

Excluded:

- Direct vending machine automation
- Payment integration

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9. 🚫 Out of Scope (MVP)

- UPI / online payments
- Passenger login system
- Analytics dashboard

- Inventory tracking
- Multi-language support
- Direct vending machine control

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10. ⚡ Performance Requirements

- Order latency: <1 second
- UI response: instant feedback
- System uptime: 99% during trip

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11. 🧪 Testing Requirements

- Simulate 30-seat usage
- Test:
    - Multiple simultaneous orders
    - Network delays
    - UI responsiveness

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12. 🚀 Deployment Plan

Phase 1:

- Develop UI + backend
- Internal testing

Phase 2:

- Deploy in 1 KSRTC bus
- Real-world trial

Phase 3:

- Iterate based on feedback

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13. 📊 Success Metrics

- Orders per trip

- Avg delivery time
- Passenger usage rate (%)
- System uptime

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14. ⚠️ Constraints

- Must work on existing bus screens
- Limited hardware power
- Intermittent internet possible
- Staff must operate quickly

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15. 🔮 Future Enhancements

- UPI payment integration
- Pre-booking system
- Snack add-ons
- Ads on screens
- AI-based recommendations

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16. 🧭 Key Principle

Keep system simple, fast, and reliable.
Avoid over-engineering in MVP stage.